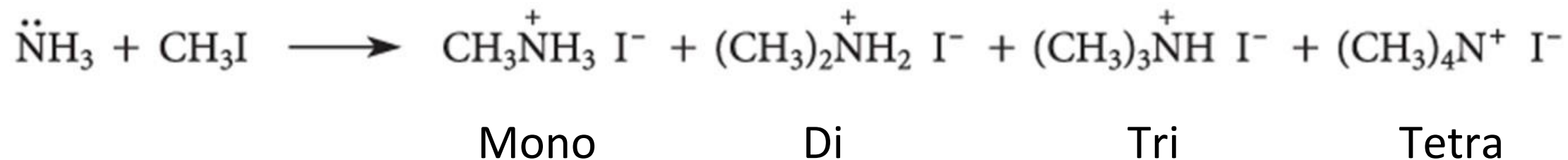


Chapter 23 (Carey Chapter 22)

The Chemistry of Amines

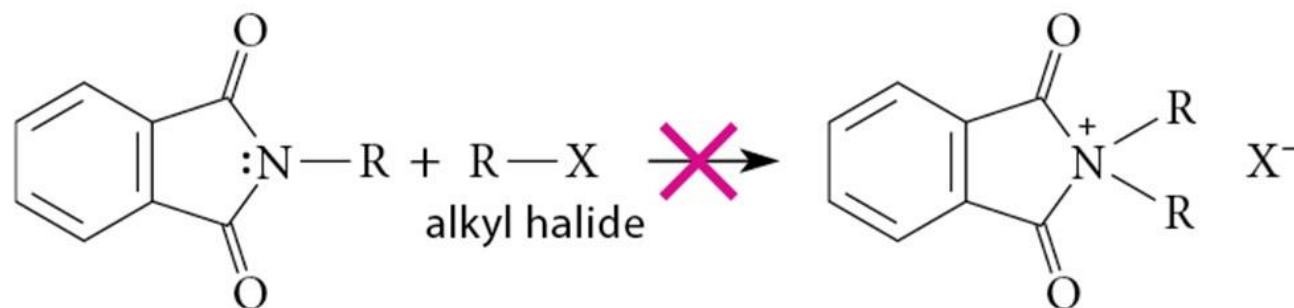
How to prepare primary amines?

- It is difficult to stop the alkylation after the first step:



Synthesis of Primary Amines

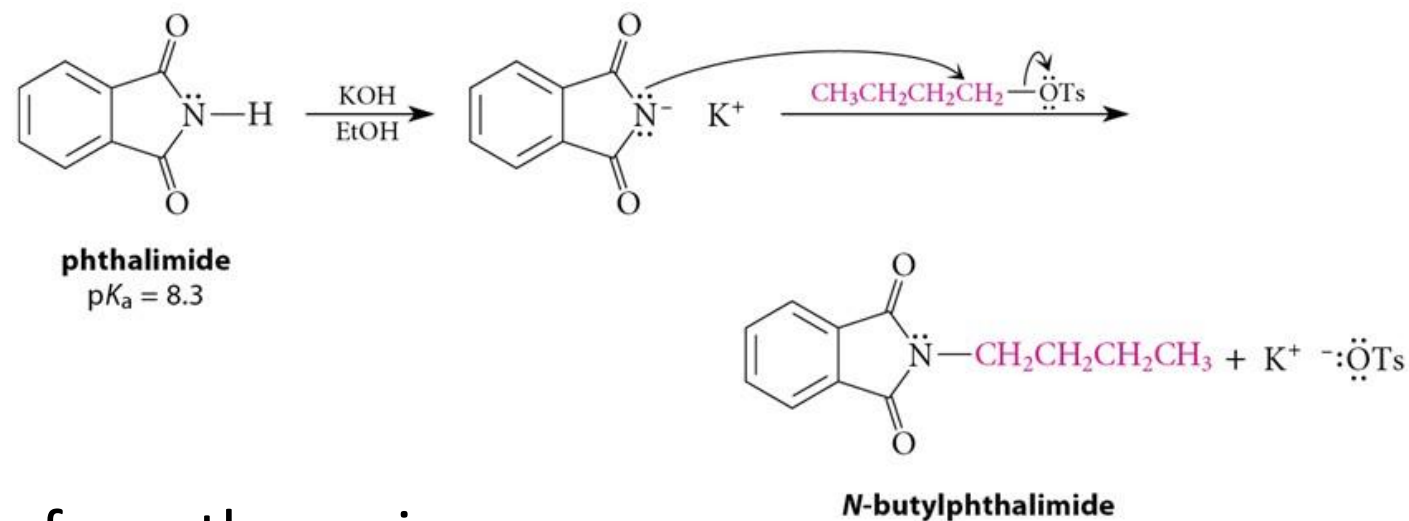
- Direct alkylation is not a good method for the preparation of primary amines.
- The **Gabriel synthesis** allows for *controlled* preparation.
 - Alkylation followed by hydrolysis
 - Single alkylation requires a protected amine group, such as phthalimide:



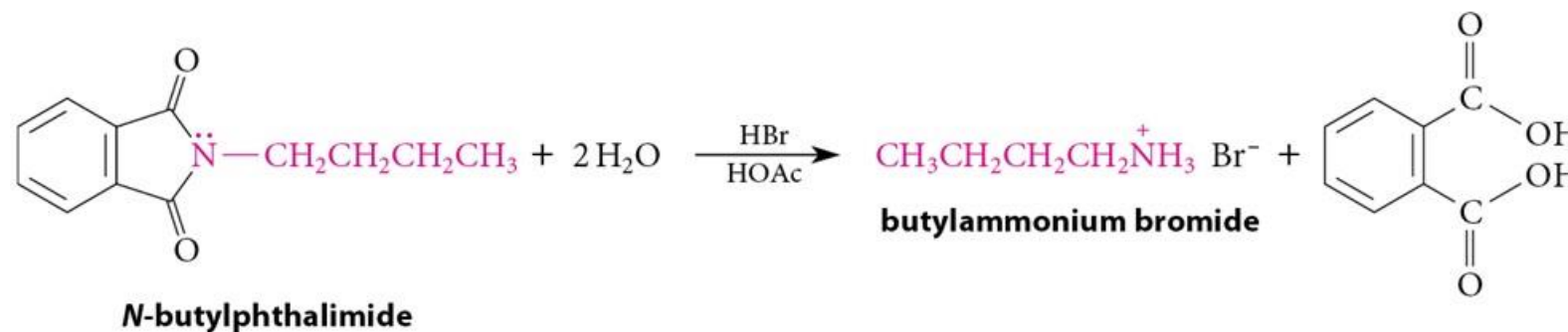
- The reason NH₃ would not be the best to make a primary amine shown on the board.

Steps in Gabriel Synthesis

- Alkylation of phthalimide



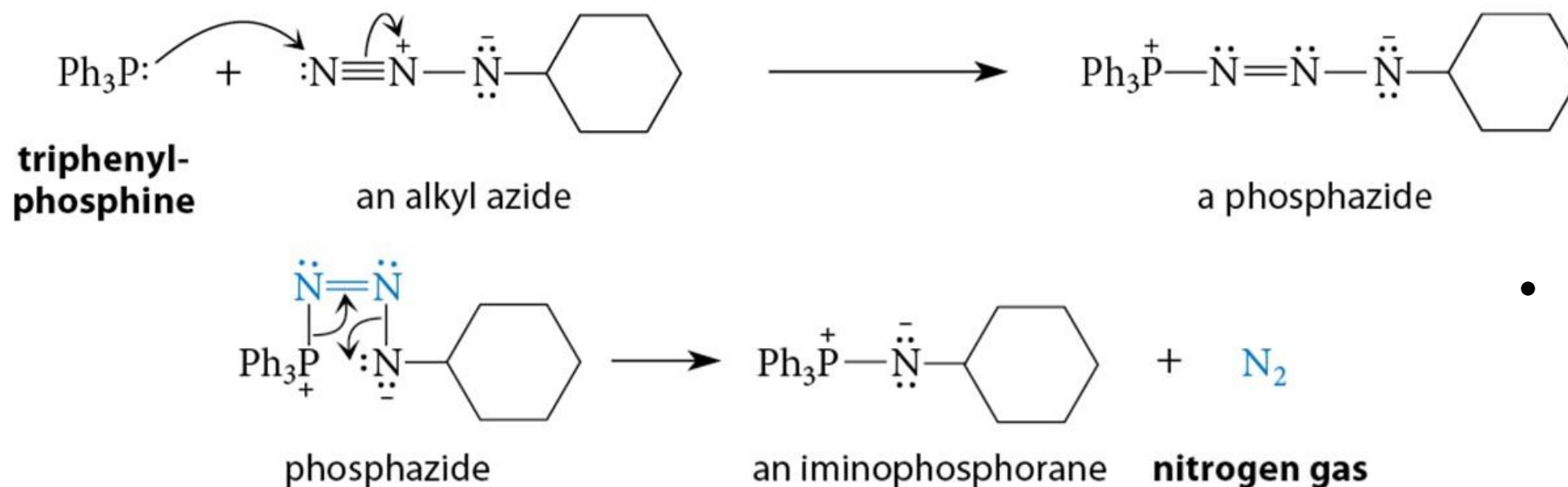
- Hydrolysis of the phthalimide frees the amine.



The Staudinger Reaction Also Avoids Multiple Alkylations

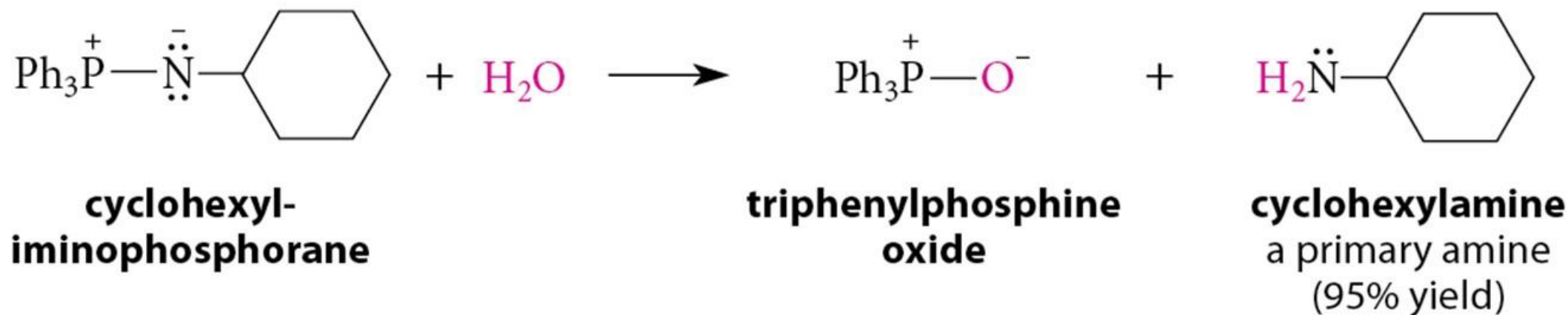


propyl azide
(88% yield)

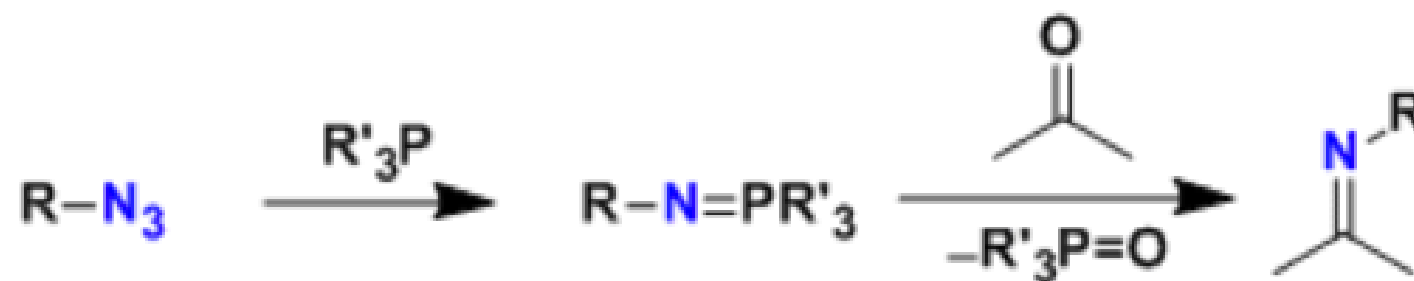


- Wittig reaction recap on the board.

The Staudinger Reaction: Hydrolysis of Iminophosphorane



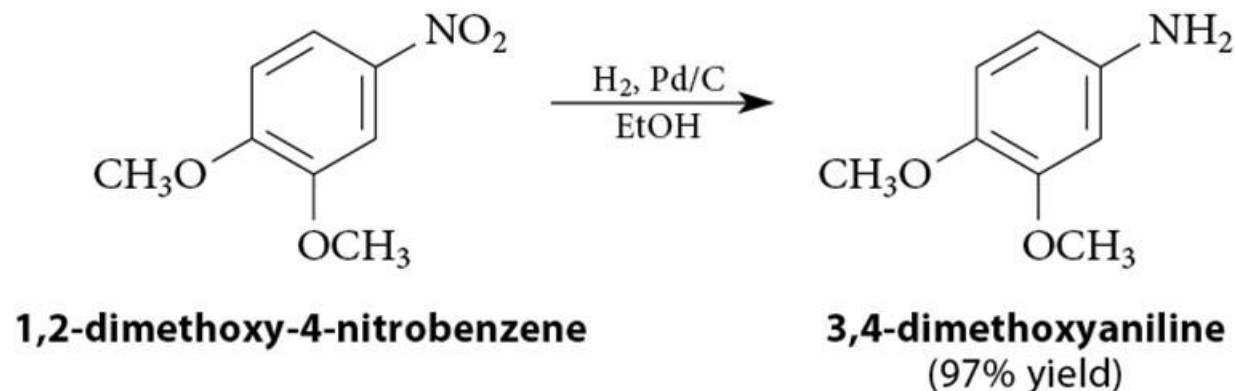
Aza-Wittig reaction



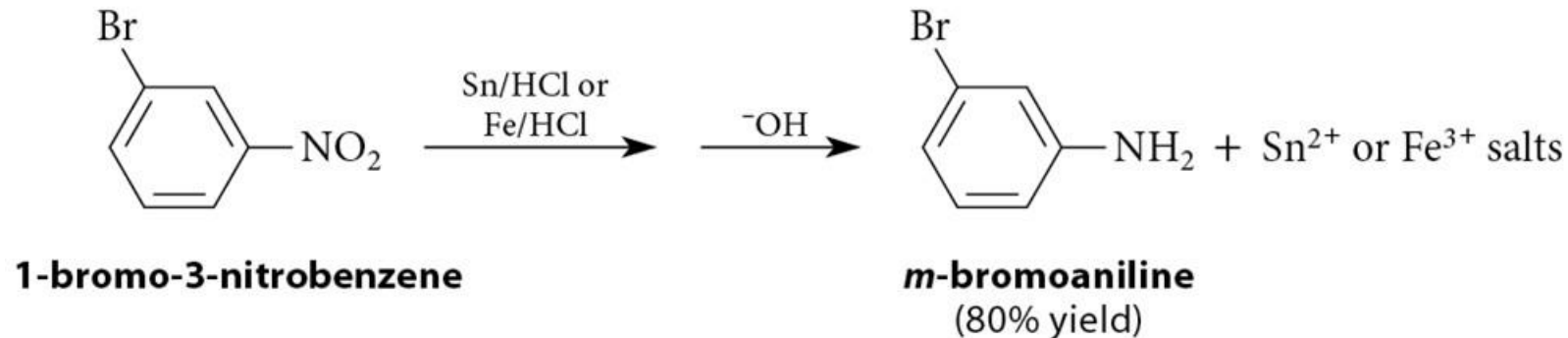
<https://en.chem-station.com/reactions-2/2015/03/aza-wittig-reaction.html>

Reduction of Nitro Compounds

- Catalytic hydrogenation:



- Reduction with finely divided metal powders:



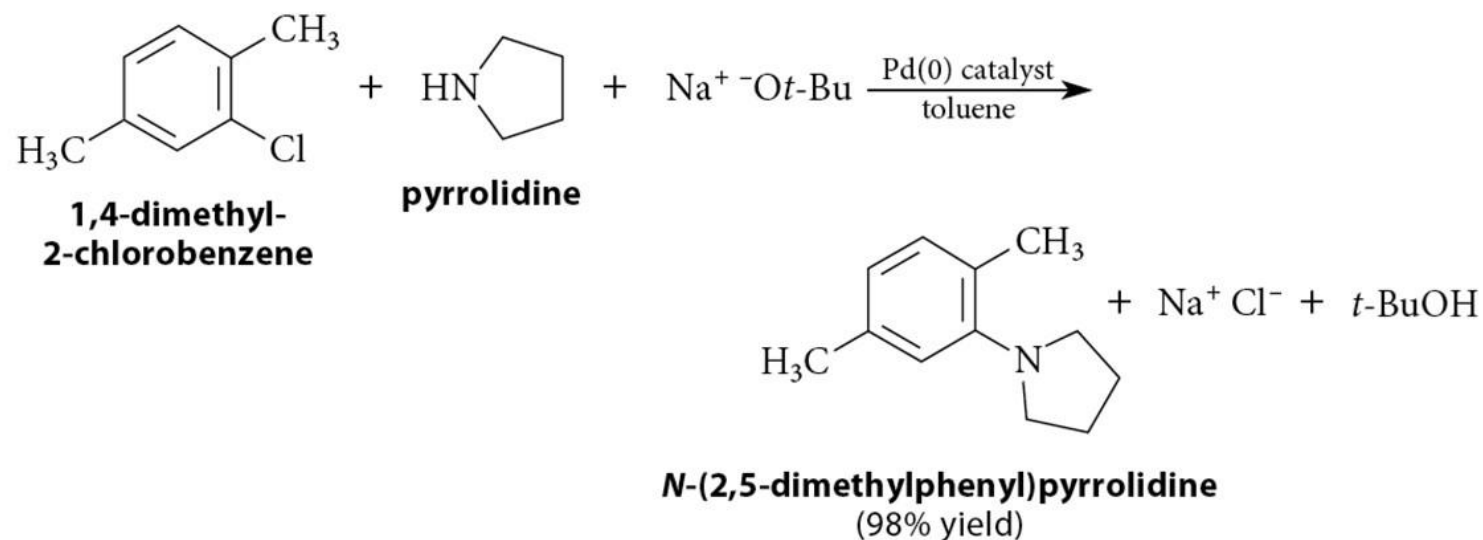
Synthesis of aniline by nitro reduction

- Details on the board.
 - 1) Nitration of benzene
 - 2) Reduction of nitro group to amine (aniline)

The directing effect difference between aniline and nitro on the board as well.

Amination of Aryl Halides

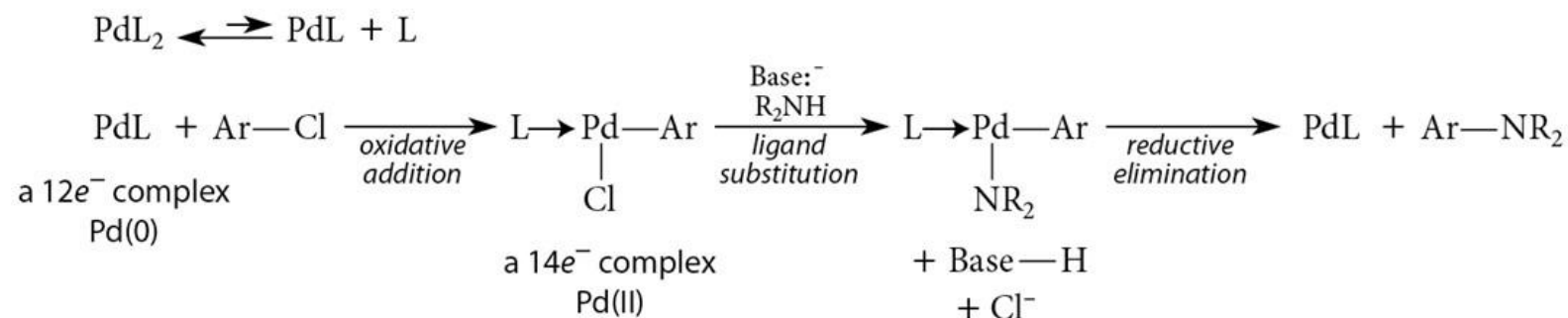
- Direct amination of aryl halides requires a base and a Pd(0) catalyst:
(Buchwald–Hartwig amination)



- Cross coupling recap on the board.

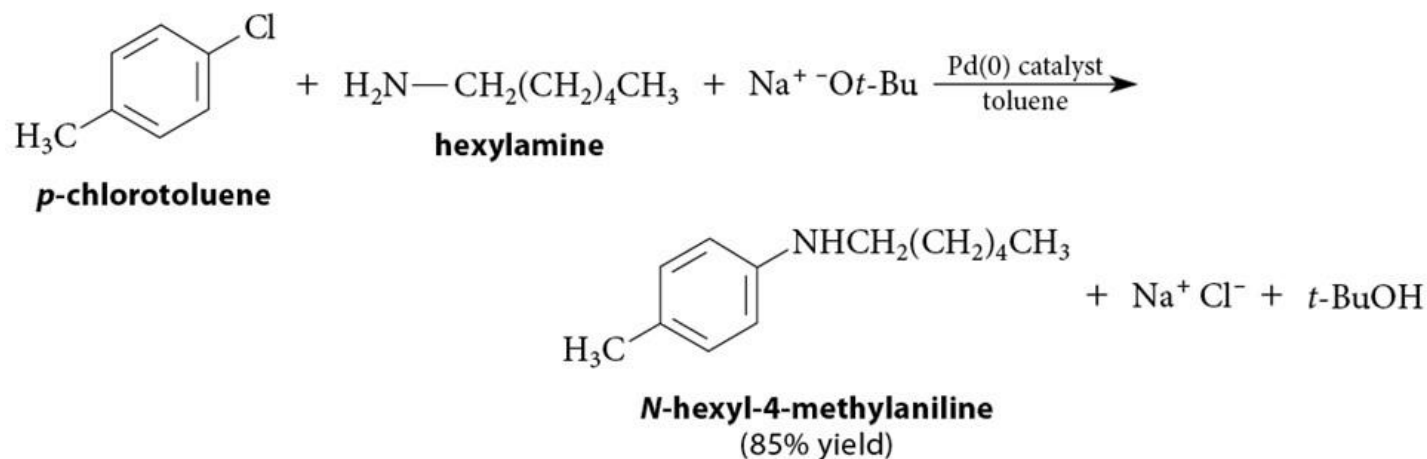
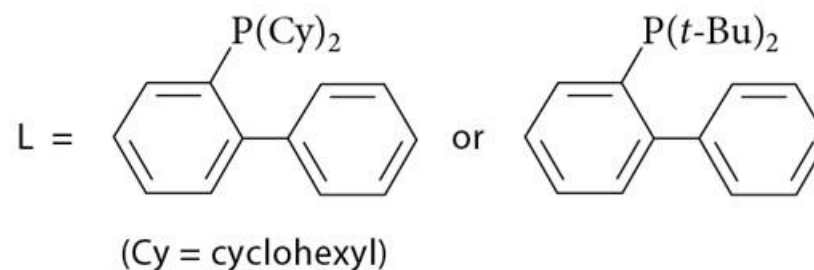
Buchwald–Hartwig Amination

- Key steps =
oxidative addition
and reductive
elimination
- Subsequent
aminations possible
with primary
amines



Minimize Second Amination with Catalyst

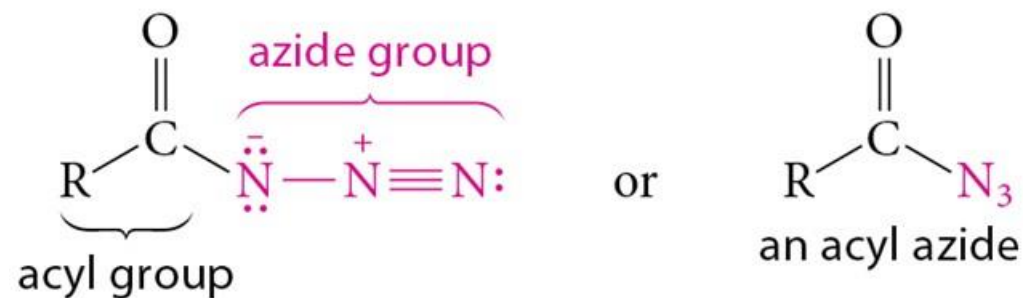
- Catalyst with sterically demanding ligands impedes second amination:



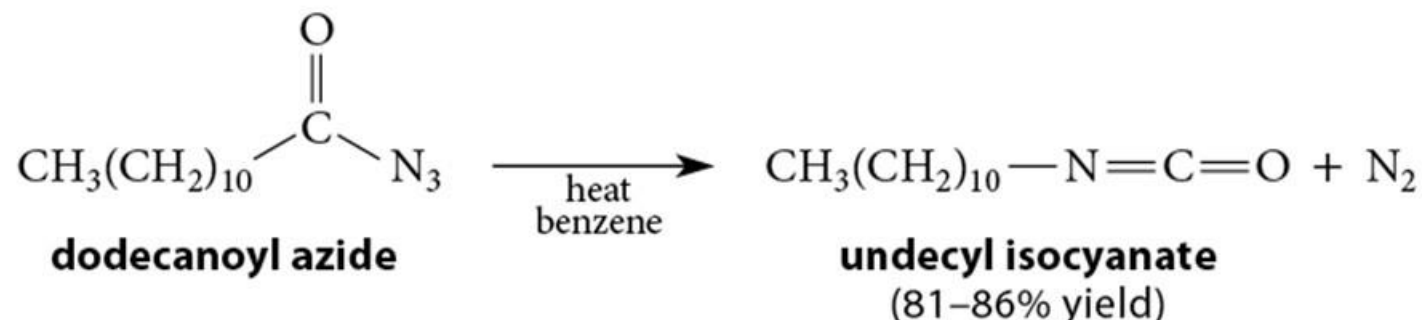
The following slides are just for your references.

The Curtius Rearrangement (1 of 2). *This one is advanced..!*

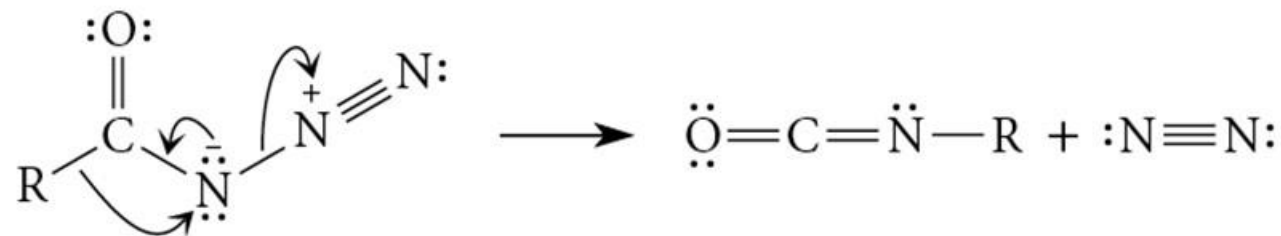
- **Acyl azide** reactant:



- Heat in inert solvent to form **isocyanates**.

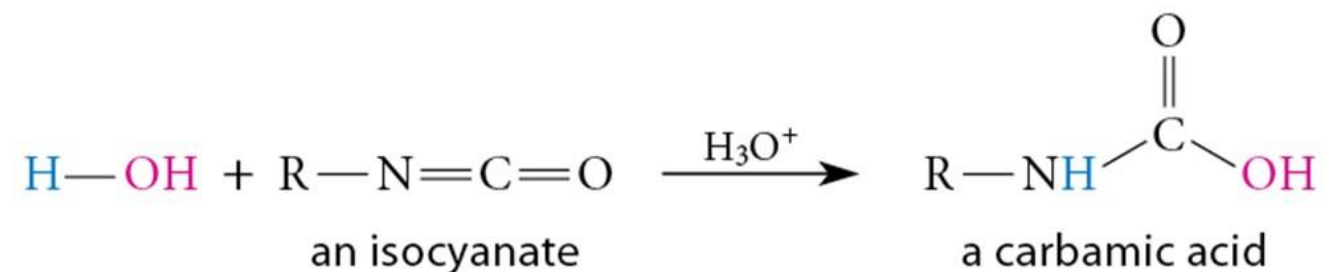


- Concerted mechanism:

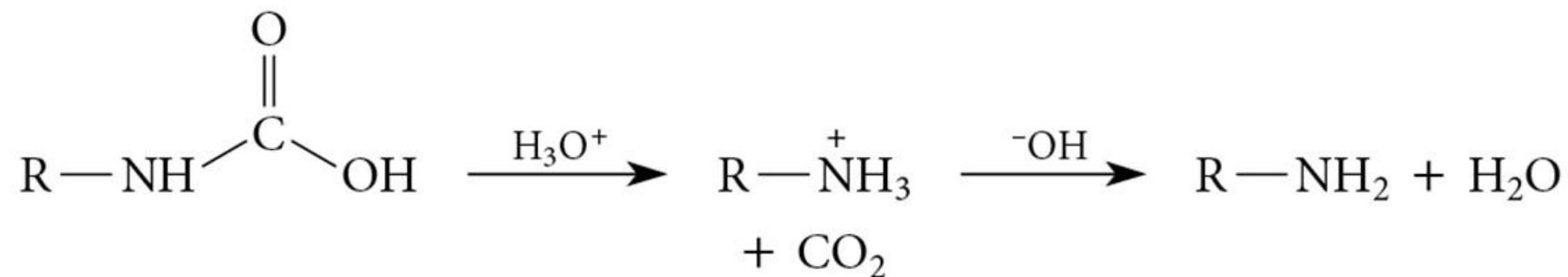


The Curtius Rearrangement (2 of 2)

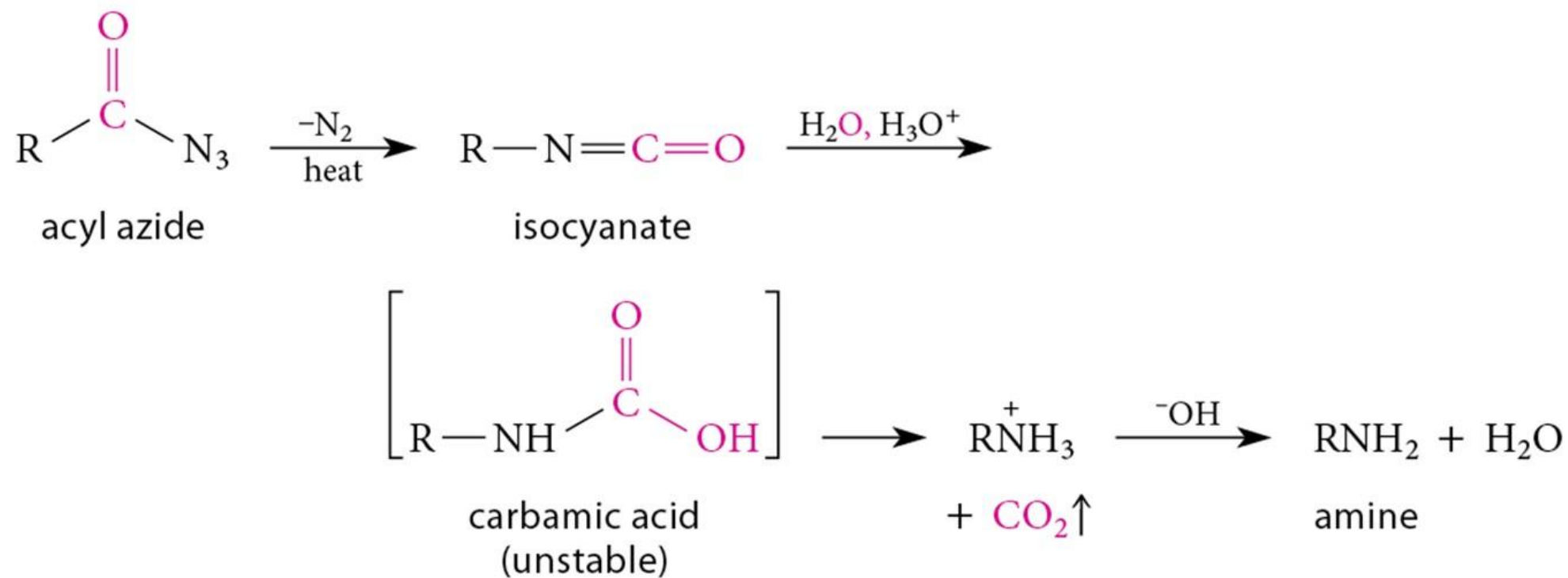
- The resulting isocyanate can be hydrated in acid or base.



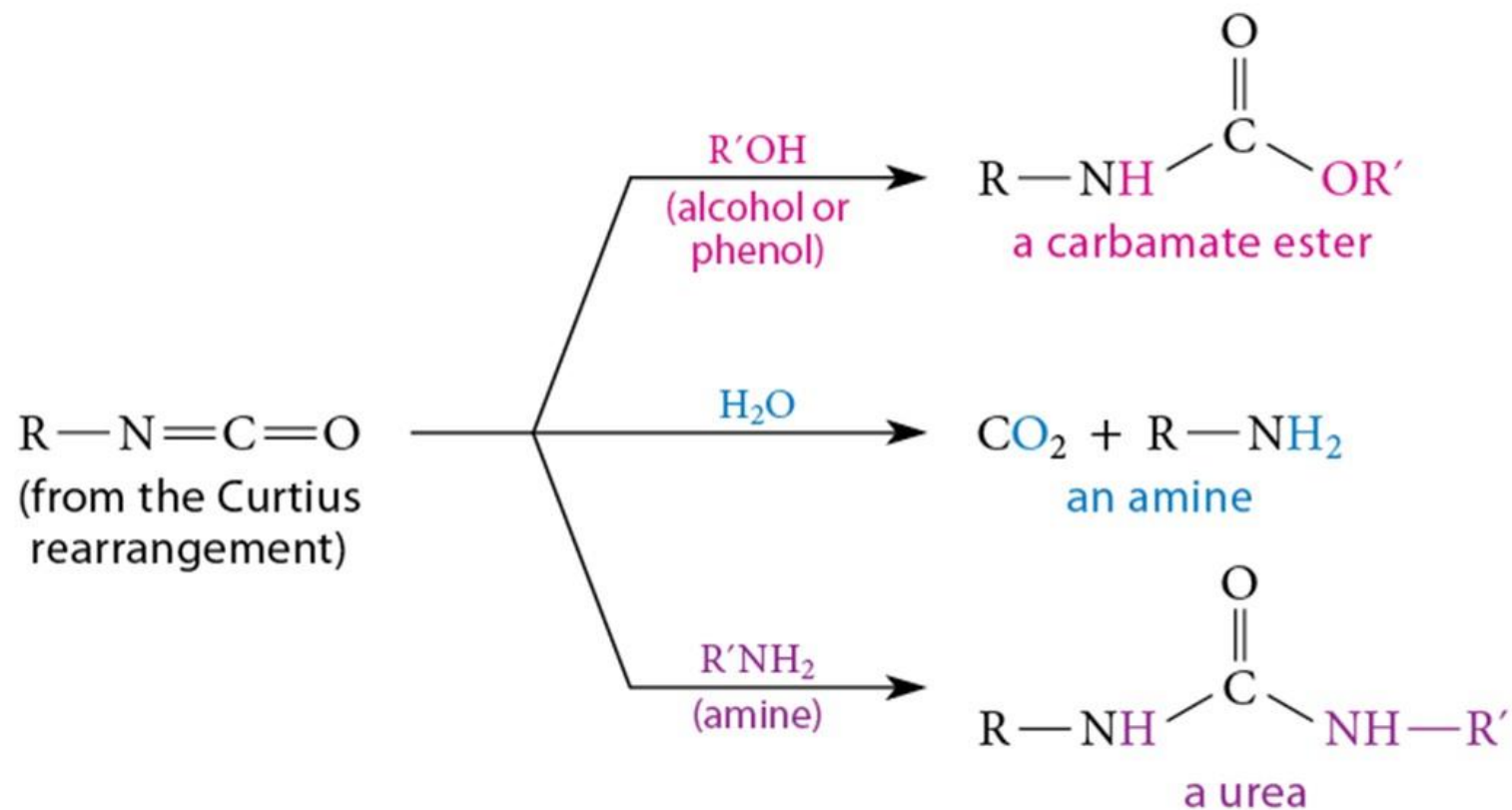
- Carbamic acids spontaneously decarboxylate.



The Curtius Rearrangement: Overall Transformation

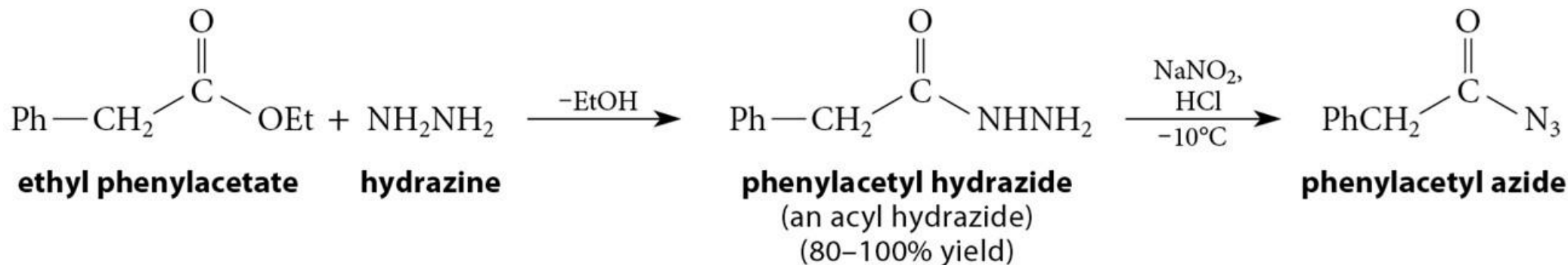
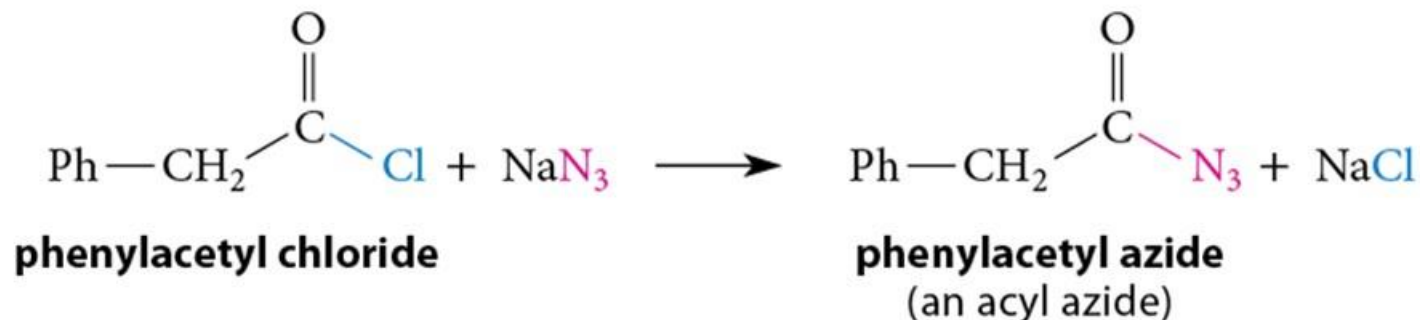


Uses of Isocyanates

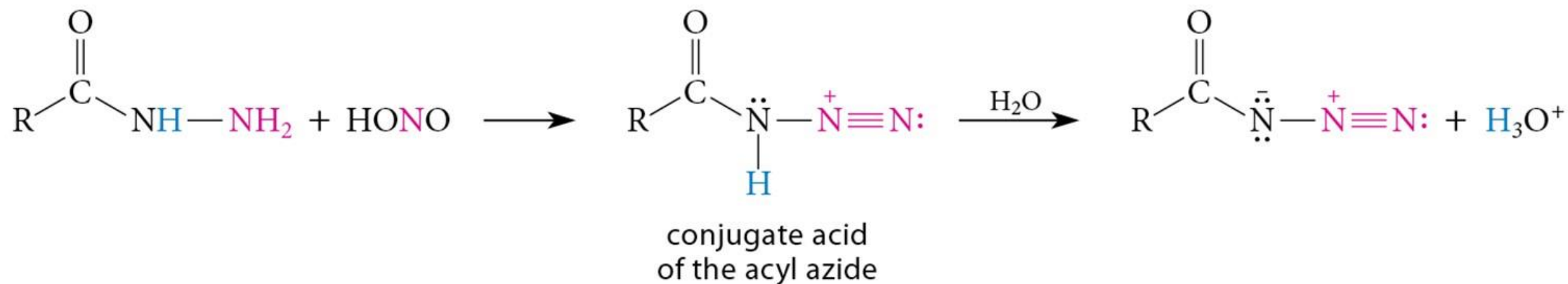
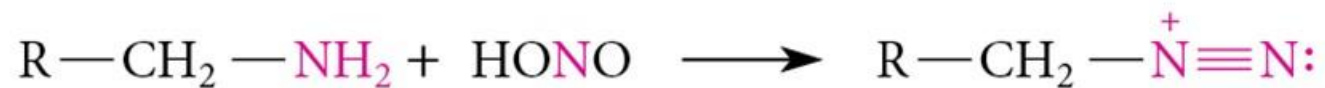


Preparation of Acyl Azides for the Curtius Rearrangement

- The acyl azides can be prepared from carboxylic acid derivatives.



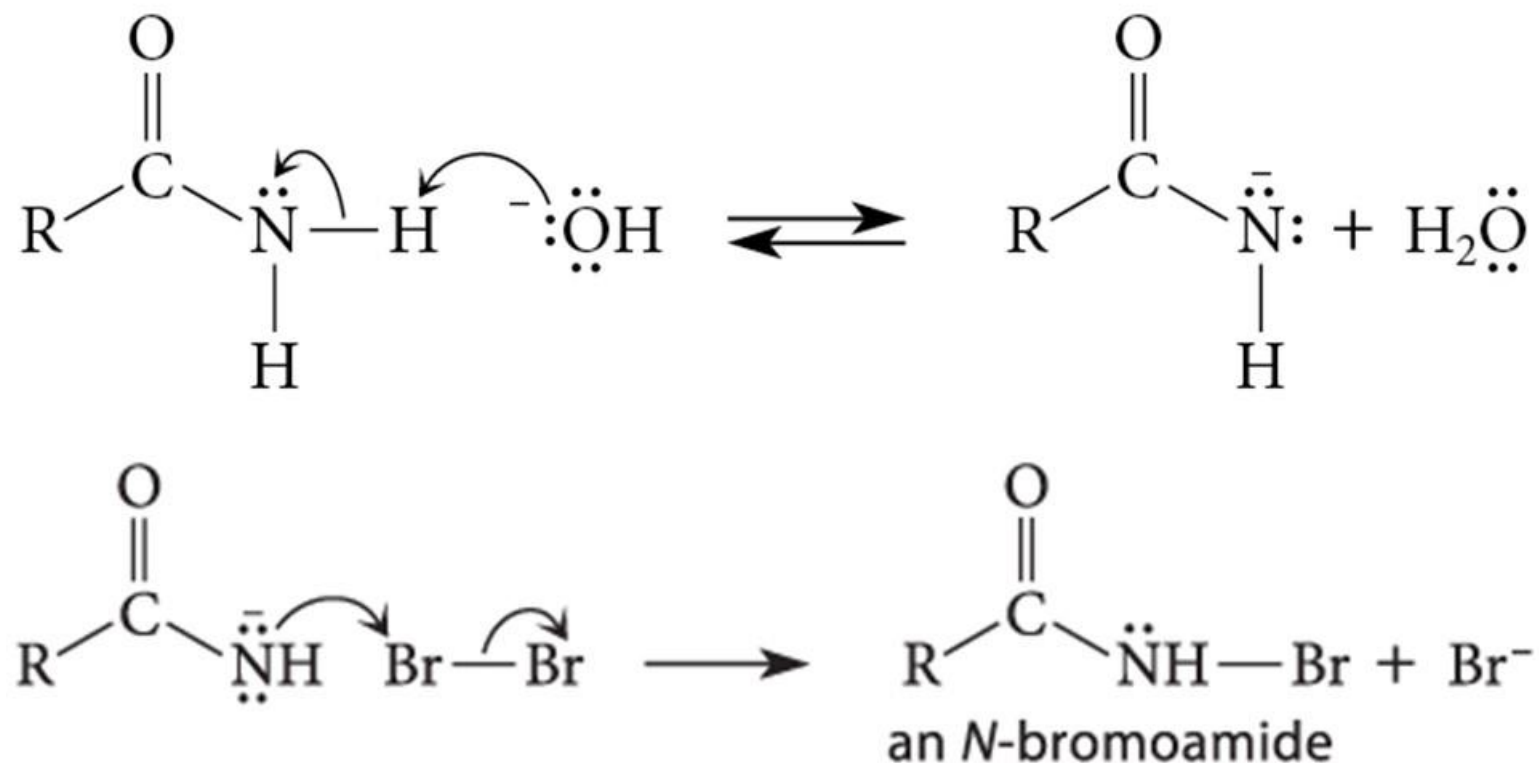
Compare Diazotization Reactions: Acyl and Alkyl Amines



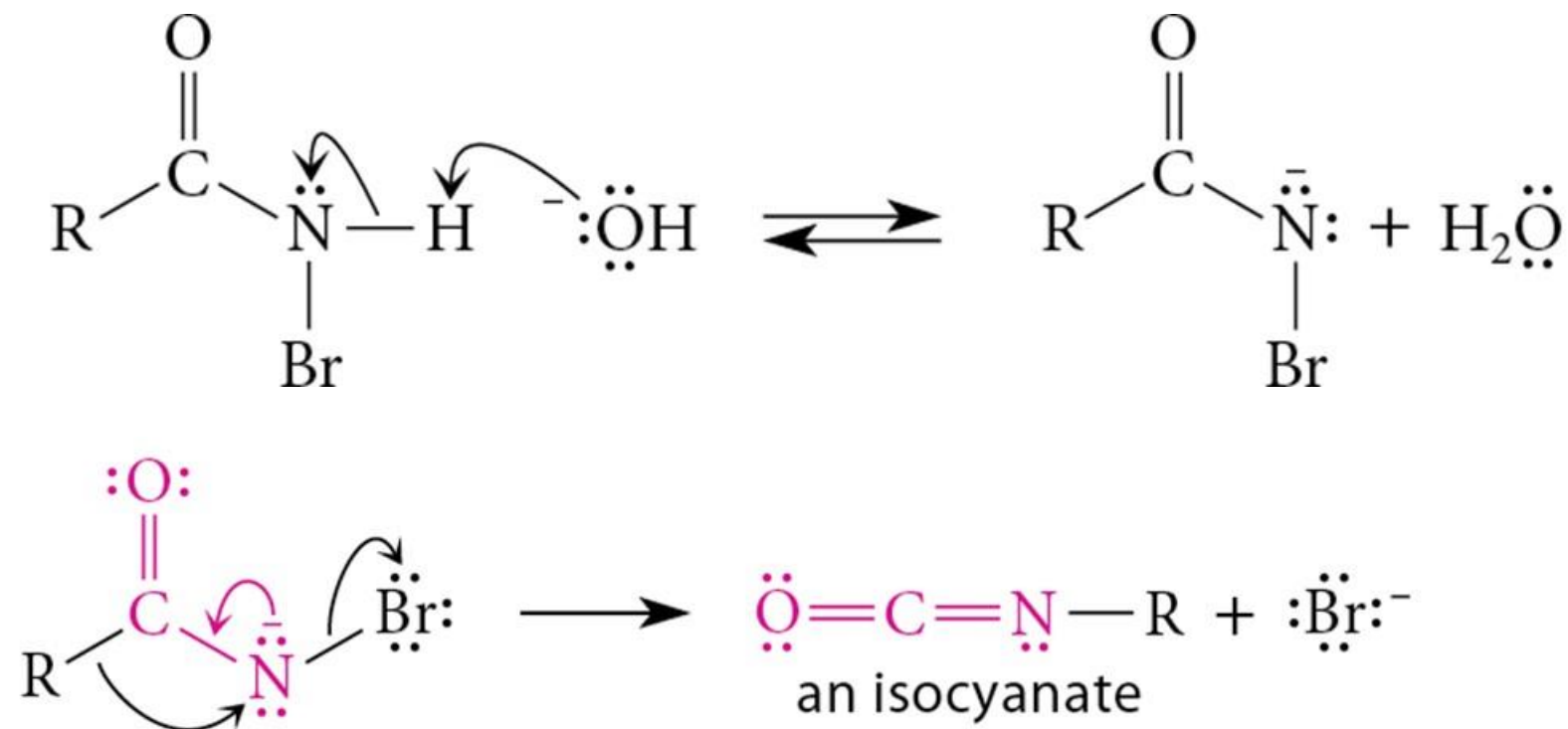
***This one is
advanced too..!***

- $$\text{Br}_2 + 2 \text{NaOH} + (\text{CH}_3)_3\text{CCH}_2\overset{\text{O}}{\parallel}\text{C}\text{NH}_2 \longrightarrow (\text{CH}_3)_3\text{CCH}_2\text{—NH}_2 + \text{O}=\text{C}=\text{O} + 2 \text{NaBr} + \text{H}_2\text{O}$$
- 3,3-dimethylbutanamide** **2,2-dimethyl-1-propanamine
(neopentylamine)**

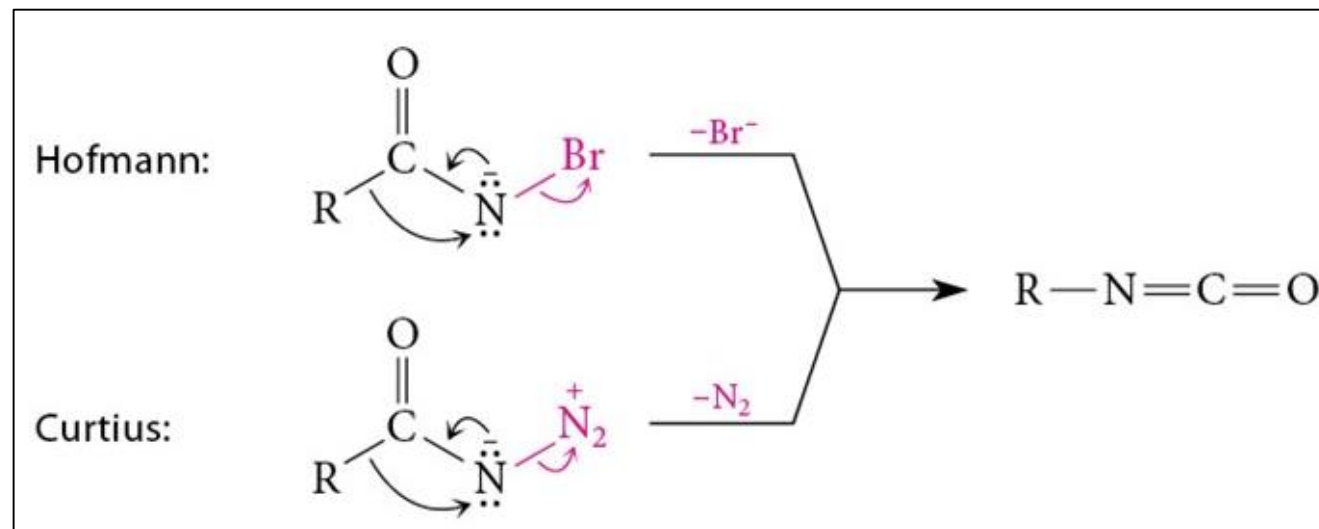
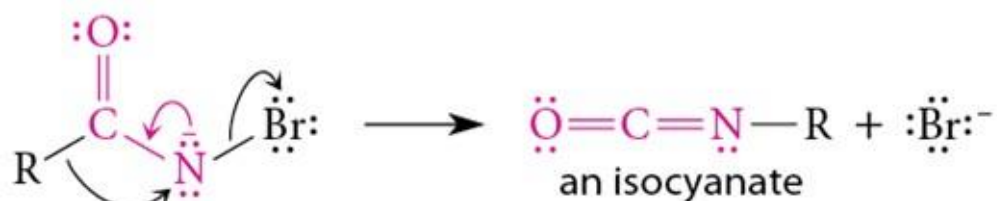
Steps in the Hofmann Rearrangement (1 of 2)



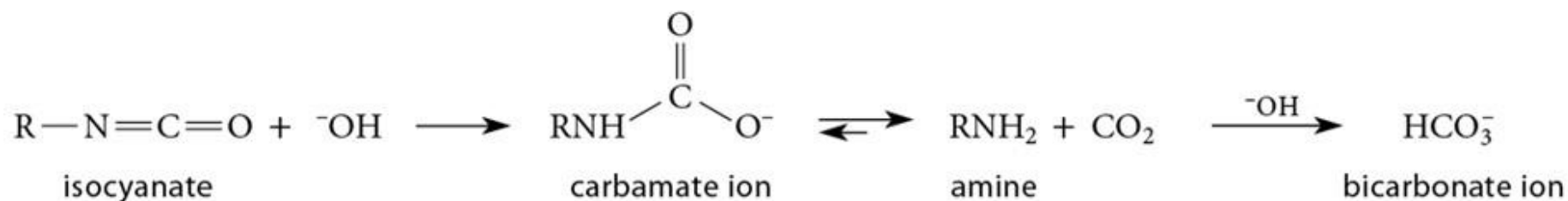
Steps in the Hofmann Rearrangement (2 of 2)



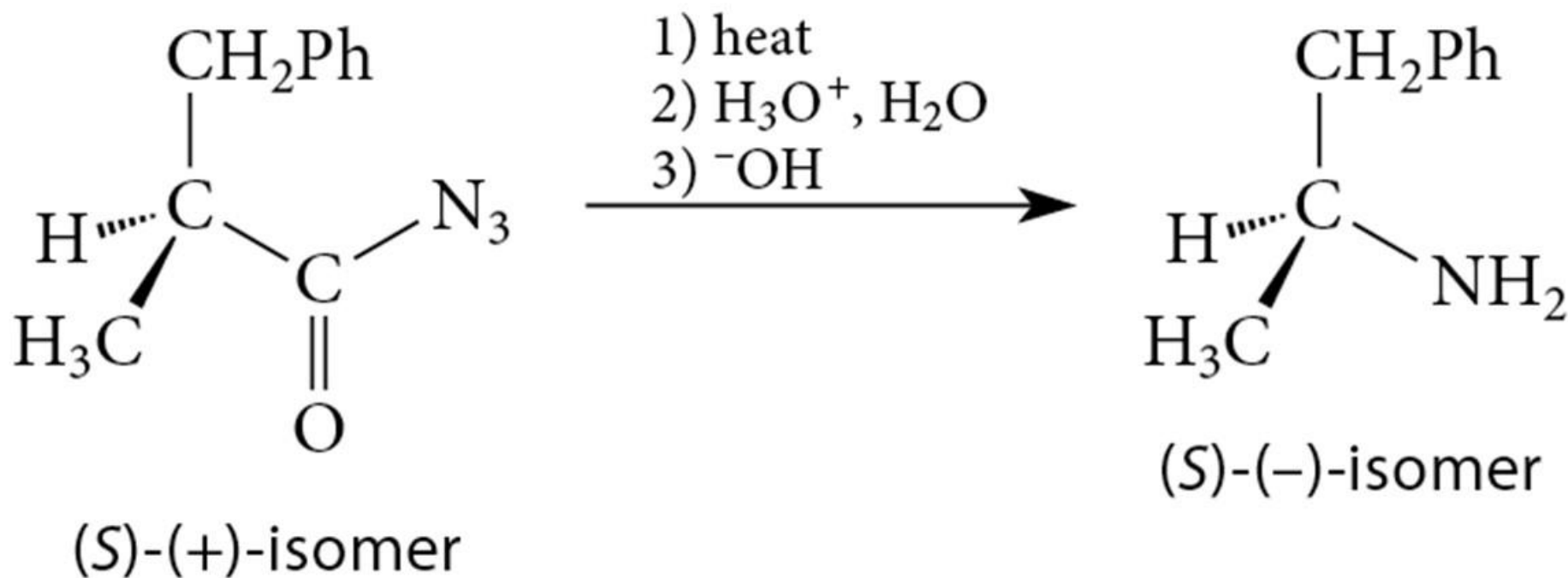
Comparing Hofmann and Curtius Rearrangements



- In the Hofmann rearrangement, the isocyanate cannot be isolated as the reaction is carried out in aqueous base.



Both Rearrangements: Alkyl Group Migrates with Complete Retention of Configuration



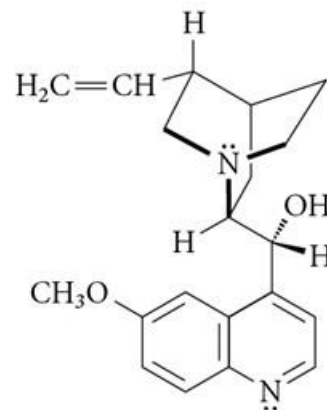
Summary of Amine Syntheses

1. Reduction of amides and nitriles with LiAlH_4
2. Direct alkylation of amines (limited in use)
3. Reductive amination
4. Aromatic substitution reactions of anilines
5. Direct amination of aryl halides
6. Gabriel synthesis (primary amines)
7. Staudinger reaction
8. Reduction of nitro compounds
9. Hofmann and Curtius rearrangements

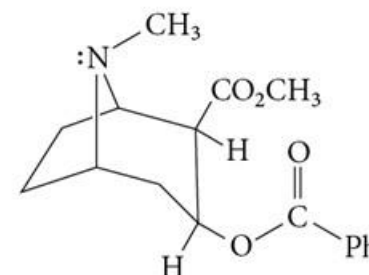
Industrial Use of Amines and Ammonia

- Hexamethylenediamine, $\text{H}_2\text{N}(\text{CH}_2)_6\text{NH}_2$ is an industrially important amine used in the synthesis of nylon-6,6.
- Ammonia, NH_3 , is a key source of nitrogen in many manufacturing processes.
- In agriculture, NH_3 and urea are important fertilizers.

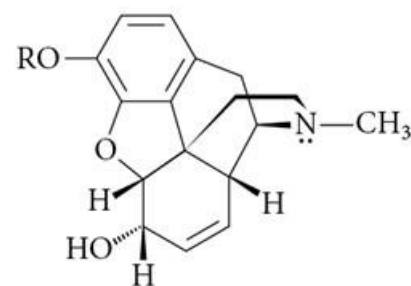
Naturally Occurring Amines (Plants): Alkaloids



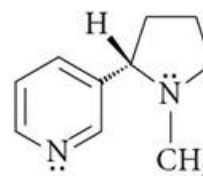
quinine
(an antimalarial drug)



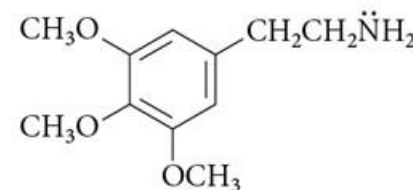
cocaine
(a stimulant of the central nervous system;
induces euphoria; widely abused)



morphine (R = H)
codeine (R = CH₃)
(medically important analgesics)



nicotine
(the principal alkaloid
from tobacco)



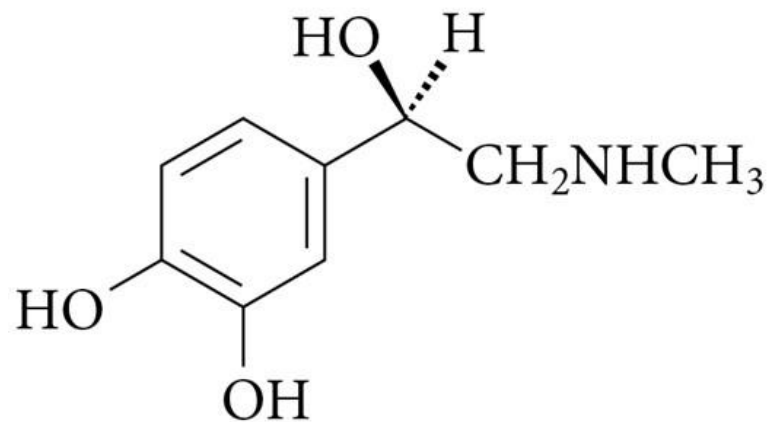
mescaline
(a hallucinogen from
peyote cactus)

Loudon/Parise, Organic Chemistry, 7e, © 2021 W. H. Freeman and Company

23.12 Use and Occurrence of Amines

Naturally Occurring Amines: Hormones

- **Hormones:** compounds that regulates the biochemistry of multicellular organisms, particularly vertebrates.



epinephrine

Naturally Occurring Amines: Neurotransmitters

- **Neurotransmitters:** Molecules involved in the communication between nerve cells, or between nerve cells and their target organs.

